# Siva Sankar Udaya Kumar

Trondheim, NO | +47-94712411 | Portfolio | Video CV | cvakapoor@gmail.com | LinkedIn

AI & ML Engineer with a strong software engineering background, experienced in building **production scale AI systems, data pipelines and analytics solutions**. Specialized in **GenAI, financial & predictive analytics and decision support**, with hands-on experience delivering **secure, observable services** using **Azure, AWS and modern MLOps practices**.

## Experience

### Freelance Projects — Dec 2024 – Present

*GenAI Credit Risk Agent* — *Remote*

- Built a production grade ***GenAI agent*** using ***ASP.NET Core*** and ***Azure OpenAI*** for financial document and credit risk analysis.
- Implemented ***Retrieval Augmented Generation (RAG)*** with ***Azure AI Search*** and ***MCP style agent orchestration*** to ensure grounded, explainable outputs.
- Developed ***secure REST APIs*** with role based access control, structured logging, audit trails and observability to support safe LLM usage in regulated environments.

*Real-Time Stock Analytics Pipeline* — *Remote*

- Developed a real-time data pipeline using ***Kafka, Airflow, MinIO***, and ***PostgreSQL*** to ingest, store, and process live stock market data from the Finnhub API.
- Designed ***Airflow DAG***s to compute volatility metrics (log returns, Sharpe ratio) and trigger alerts for anomalies such as price drops or abnormal volatility.
- Built a ***Streamlit*** dashboard for real-time monitoring with interactive visualizations of stock trends and alert events.

### Norwegian University of Science and Technology — Oct 2021 – Nov 2024

*PhD Candidate* — *Trondheim, Norway*

- Designed and implemented ***stochastic optimization models*** in ***Python*** to support operational decision making under uncertainty in ***large scale energy systems***.
- Developed ***scalable scenario generation pipelines*** using probabilistic modeling, ***GAN and Reinforcement Learning***, delivering reproducible and performance optimized components for decision support workflows.

### Indian Institute of Technology, Bombay — Jul 2021 – Sep 2021

*AI Researcher* — *Mumbai, India*

- Developed an ***end-to-end vehicle insurance cross-sell prediction system*** using ***Databricks, Apache Spark***, and ***XGBoost***, with MLflow for experiment tracking and model performance optimization.
- Designed and deployed a ***scalable REST API*** for model inference, implementing ***MLOps practices*** with ***Docker*** and ***Kubernetes*** for automated deployment, monitoring, and scalability.

### Relatas — Jan 2021 – June 2021

*Data Scientist* — *Bangalore, India*

- Developed a ***Smart Opportunity Recommender*** tool using ***Python*** and ***scikit-learn***, incorporating ***sentiment analysis*** and ***NLP*** to identify high potential sales leads.
- Deployed models on ***AWS*** for scalability and performance, managing workflows with ***Jira*** in an ***agile*** environment.

### Iha Consulting Services — Oct 2020 – Dec 2020

*ML Engineer* — *Secunderabad, India*

- Developed a ***Human Activity Recognition model*** using Ensemble Learning on multi sensor time series data, performing feature engineering and exploratory analysis to enhance predictive accuracy.
- Built an ***end-to-end MLOps pipeline*** for model training, versioning, deployment, and monitoring, integrating ***CI/CD workflows*** for production grade reliability.

### iSmile Technologies — June 2020 – Sep 2020

*AI Engineer* — *Bolingbrook, United States (Remote)*

- Led a ***team of three*** in integrating Computer Vision and Robotics for autonomous object detection using a custom model built with ***Azure Custom Vision***.
- Developed an ***Android app*** for real-time detection and coordinated delivery using ***Azure DevOps*** and ***SAFe***, ensuring streamlined collaboration and project execution.

### CodeSpeedy Technology — Jan 2020 – June 2020

*AI Developer* — *Kolkata, India (Remote)*

- Built and deployed ***end-to-end ML systems*** across ***NLP, Computer Vision and Time Series*** using ***Python, TensorFlow*** and ***AWS/Azure***.
- Operationalized models with ***CI/CD, observability*** and ***scalable API based serving*** in production environments.

## Technical Skills

**Programming Languages**: Python, SQL, C# (.NET), MATLAB, Julia
**GenAI & ML Systems**: Azure OpenAI, LLMs, RAG, Agent Orchestration (MCP), REST API
**Cloud & Infra**: Databricks, Azure Cloud, GCP, AWS, Docker, Kubernetes, Terraform
**Big Data & Streaming**: Apache Spark, Kafka, Airflow, PostgreSQL, MinIO, Hive, BigQuery
**MLOps & Observability**: MLflow, CI/CD, Git, GitLab, Jira
**ML & AI Frameworks**: TensorFlow, Scikit-learn, PyTorch, XGBoost
**Dashboarding & Spreadsheet**: PowerBI, Streamlit, Data Studio, Tableau, MS Excel

## Education

**Norwegian University of Science and Technology** — Oct 2021 (ABD)
*Ph.D. in Industrial Economics and Technology Management* — *Trondheim, Norway*
- Thesis: Short-Term Optimization under Uncertainty in the Norwegian Natural Gas System

**Indian Institute of Technology (BHU), Varanasi** — Jul 2019 – June 2021
*M.Tech. in Industrial Management (GPA: 9.11)* — *Varanasi, India*
- Thesis: Automated Detection and Tracking of Sewer Pipe Problems Using Inspection Videos

**Jain University** — Jul 2015 – June 2019
*B.Tech. in Aerospace Engineering (GPA: 8.01)* — *Bangalore, India*

## Other Experience

**McDonald's** — April 2026 – Present
*Crew Member (Part Time)* — *Trondheim, Norway*

**Pearl of India Restaurant** — Mar 2026 – Present
*Service Staff (On Call)* — *Trondheim, Norway*
- Customer service in a fast paced environment
- Team collaboration and shift based work
- Order handling and quality assurance

## Spoken Languages

Tamil, English, Norwegian, Hindi, Telugu, Kannada